# Pandas Data Analysis

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# 📄 Workout Data Analysis using Pandas

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## 🟢 1. Introduction

This project analyzes a workout dataset containing information such as duration, pulse rate, maximum pulse, and calories burned. The aim is to understand how different workout factors affect calorie burn and overall performance.

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## 🟡 2. Objective

The objective of this project is:

- To analyze the relationship between workout duration, pulse rate, and calories burned
- To identify patterns in high-intensity workouts
- To determine whether duration or intensity has a greater impact on calorie burn

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## 🔵 3. Data Cleaning

The dataset contained some missing values in the "Calories" column. These missing values were handled by replacing them with the mean value of the column. This ensures that the dataset remains consistent and usable for analysis.

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## 🟣 4. Data Analysis

The analysis was performed using Pandas library in Python. The following operations were carried out:

- Grouping data based on workout duration to calculate average calories burned
- Finding maximum pulse values for different durations

- Performing correlation analysis to identify relationships between variables
- Filtering high-intensity workouts based on pulse and calories
- Sorting data to identify top calorie-burning workouts

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## 🔴 5. Insights

### 🔹 Duration vs Calories

Short workouts (15–25 minutes) show an increase in calories burned. However, for longer durations (30–45 minutes), the average calories burned becomes relatively stable. This indicates that increasing duration alone does not significantly increase calorie burn.

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### 🔹 Pulse vs Calories

The correlation between pulse and calories is weakly positive, meaning that higher pulse slightly increases calorie burn. However, maximum pulse shows a stronger relationship with calorie efficiency, indicating that peak heart rate plays a more important role.

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### 🔹 Importance of Intensity

High-intensity workouts (pulse > 110) are associated with significantly higher calorie burn. These workouts are more effective compared to lower-intensity sessions, even if the duration is similar.

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### 🔹 High Intensity Workouts

The filtered high-intensity data shows that workouts with higher pulse values consistently result in calories burned above 300. This highlights the importance of maintaining a higher heart rate during workouts.

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### 🔹 Peak Performance

The highest calorie burn values are observed in workouts with both high duration and high pulse rate. This suggests that combining longer duration with high intensity leads to the best results.

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### 🔹 Correlation Summary

The analysis shows that:

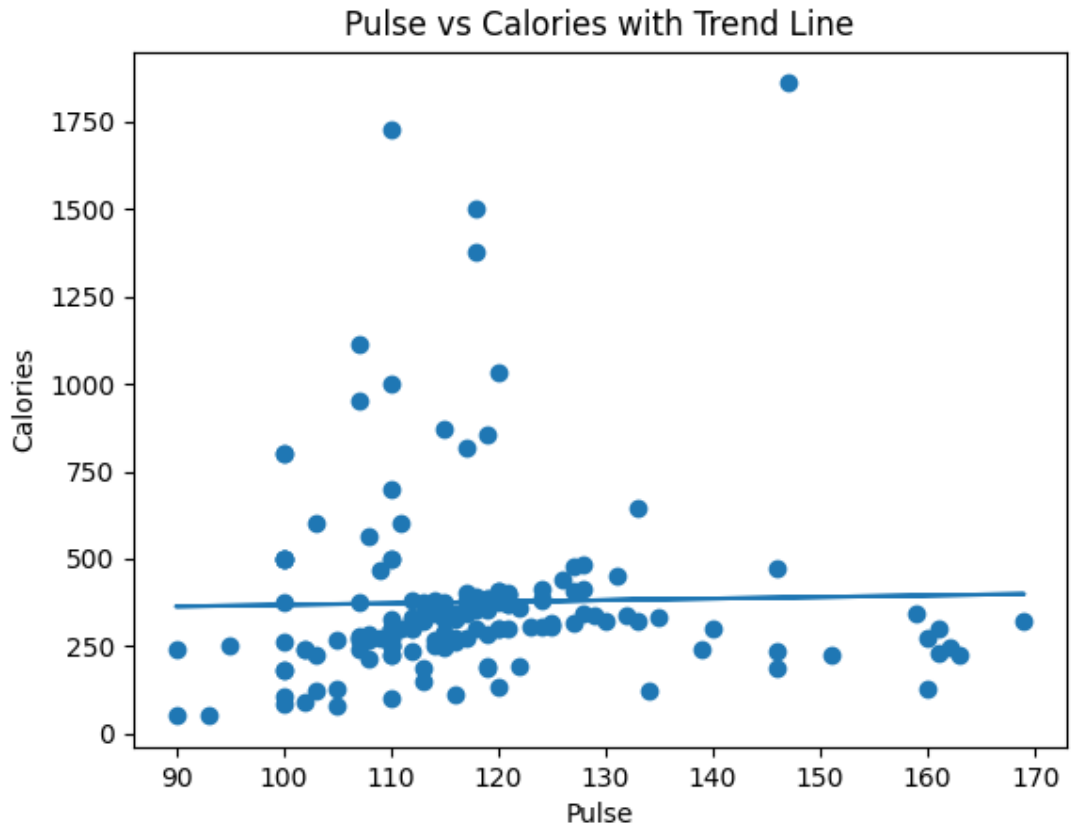
- Pulse has a weak positive relationship with calories
- Maximum pulse has a strong relationship with calorie efficiency
- Duration has a weak impact on calories

Overall, intensity has a greater influence on calorie burn than duration.

## 📄 6. Conclusion

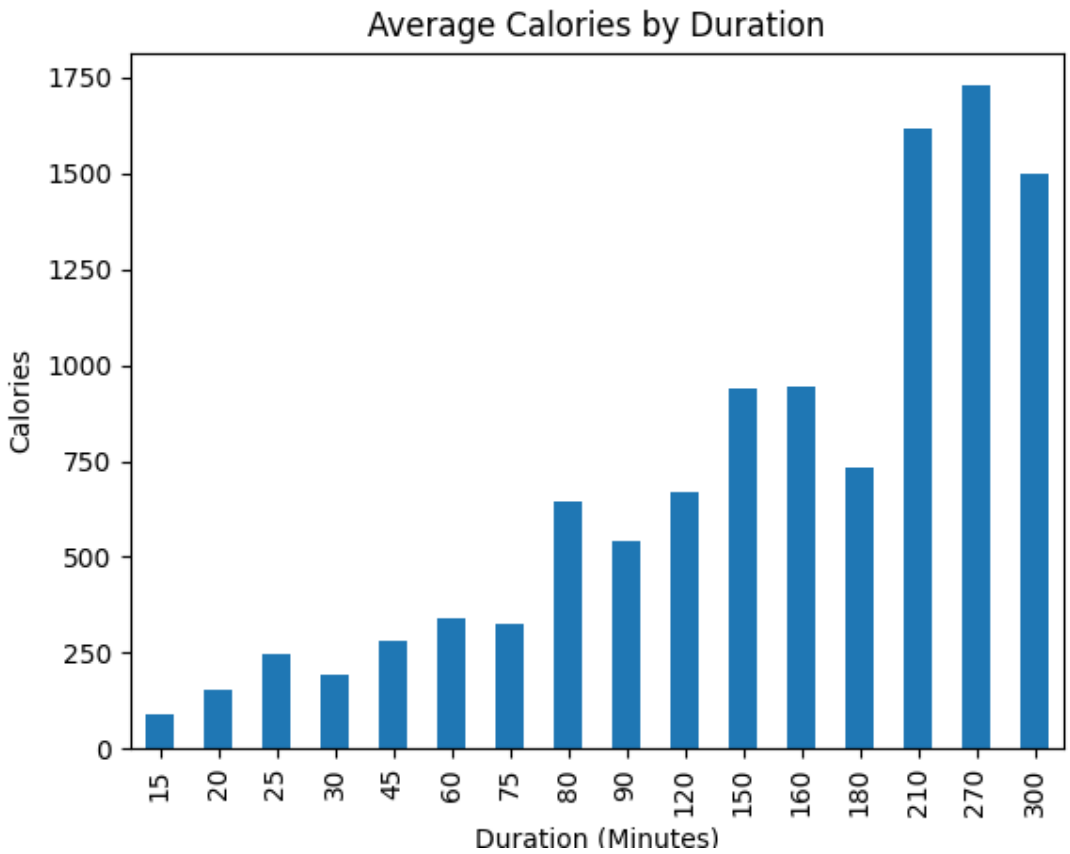
The analysis concludes that workout intensity plays a more significant role in calorie burn than duration. While longer workouts provide consistency, high-intensity sessions with higher pulse rates lead to greater calorie expenditure. Therefore, maintaining a higher heart rate during workouts is more effective than simply increasing workout duration.

### 🔹 Pulse vs Calories (Graph Explanation)




- **Pulse vs Calories (Graph Explanation)**

The scatter plot with a trend line shows the relationship between pulse rate and calories burned. It can be observed that as the pulse rate increases, the calories burned also tend to increase. The trend line indicates a positive relationship, although the correlation is not very strong. This suggests that higher intensity workouts (higher pulse) generally lead to higher calorie burn, but other factors also influence the outcome.

### 📊 Average Calories Burnt by Duration (Graph Explanation)

The bar chart shows the average calories burned for different workout durations. It can be observed that calories increase from shorter durations (15–25 minutes), but for longer durations (30–45 minutes), the average calories remain relatively stable. This indicates that simply increasing workout duration does not always lead to significantly higher calorie burn.